

Forecasting US Employment Growth

An AR(3) Approach

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Abstract

In this report, we employ an autoregressive (AR) model to analyze the dynamics of employment growth in the United States and to produce forecasts. For this purpose, we use a monthly time series of US employment covering the period 1974–2025 and construct the year-on-year growth rate in order to obtain a stationary series suitable for estimation.

1 Empirical model

1.1 Model specification

For the purpose of this study, we model employment growth rate as an autoregressive process of order three, AR(3) with a constant:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + u_t \quad (1)$$

where y_t is the observation of employment growth rate at time t (monthly frequency), and $y_{t-1}, y_{t-2}, y_{t-3}$ are respectively the first lag, the second lag and the third lag. The term c is the constant (intercept) of the model. The coefficients ϕ_1, ϕ_2, ϕ_3 are the unknown parameters we want to estimate along with the intercept c . The u_t are stochastic error terms representing shocks, that follow the same Normal distribution (white noise) and are serially uncorrelated across time: $u_t \sim \mathcal{N}(0, \sigma^2)$.

1.2 Companion Representation

Equation (1) is a linear third-order difference equation since the value of y at date t depends on its three previous values (or lags). To simplify, we can exploit the companion representation to rewrite the AR(3) as a first-order vector autoregressive equation:

$$\underbrace{\begin{bmatrix} y_t \\ y_{t-1} \\ y_{t-2} \end{bmatrix}}_{\mathbf{y}_t} = \underbrace{\begin{bmatrix} c \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{c}} + \underbrace{\begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\Phi} \underbrace{\begin{bmatrix} y_{t-1} \\ y_{t-2} \\ y_{t-3} \end{bmatrix}}_{\mathbf{y}_{t-1}} + \underbrace{\begin{bmatrix} u_t \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{u}_t} \quad (2)$$

The AR(3) can therefore be represented as a first-order vector autoregressive equation

$$\mathbf{y}_t = \mathbf{c} + \Phi \mathbf{y}_{t-1} + \mathbf{u}_t \quad (3)$$

1.3 Computing forecasts

Using the companion representation, we can rewrite the third-order difference equation into a first-order difference equation and solve it through recursive substitution. Equation (3) can be used to forecast the employment growth rate by shifting the equation h steps ahead. By shifting Equation (3) one period ahead from the last observation (y_T), we can compute the one-step ahead forecast:

$$\mathbf{y}_{T+1} = \mathbf{c} + \Phi \mathbf{y}_T + \mathbf{u}_{T+1} \quad (4)$$

The best predictor of $\mathbf{y}_{T+1}$ is its conditional expectation given the information available at time T , I_T :

$$\hat{\mathbf{y}}_{T+1|T} = E(y_{T+1}|I_T) \quad (5)$$

$$= E(\mathbf{c} + \Phi \mathbf{y}_T + \mathbf{u}_{T+1}|I_T) \quad (6)$$

By assumption, the expected value of the error term u_{T+1} is equal to zero. Hence:

$$\hat{\mathbf{y}}_{T+1|T} = \mathbf{c} + \Phi \mathbf{y}_T \quad (7)$$

In matrix form:

$$\begin{bmatrix} \hat{y}_{T+1|T} \\ y_T \\ y_{T-1} \end{bmatrix} = \begin{bmatrix} c \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_T \\ y_{T-1} \\ y_{T-2} \end{bmatrix}$$

Similarly, the two step ahead forecast can be computed through the companion representation by shifting Equation (3) two periods ahead:

$$\hat{\mathbf{y}}_{\mathbf{T}+2|\mathbf{T}} = \mathbf{c} + \Phi \hat{\mathbf{y}}_{\mathbf{T}+1|\mathbf{T}} \quad (8)$$

By replacing Equation (4) into Equation (6), we obtain:

$$\hat{\mathbf{y}}_{\mathbf{T}+2|\mathbf{T}} = \mathbf{c} + \Phi \mathbf{c} + \Phi^2 \mathbf{y}_{\mathbf{T}} \quad (9)$$

$$\begin{bmatrix} \hat{y}_{T+2|T} \\ \hat{y}_{T+1|T} \\ y_T \end{bmatrix} = \begin{bmatrix} c \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} c \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_T \\ y_{T-1} \\ y_{T-2} \end{bmatrix}$$

By focusing on the first row, we can see that this leads to:

$$\begin{aligned} \hat{y}_{T+2|T} &= c + \phi_1 c + (\phi_1^2 + \phi_2) y_T + (\phi_1 \phi_2 + \phi_3) y_{T-1} + (\phi_1 \phi_3) y_{T-2} \\ &= c(1 + \phi_1) + (\phi_1^2 + \phi_2) y_T + (\phi_1 \phi_2 + \phi_3) y_{T-1} + (\phi_1 \phi_3) y_{T-2} \end{aligned}$$

2 Data

2.1 Data Analysis

The monthly US employment data are obtained from the Federal Reserve Bank of St. Louis' FRED database. The series used is Total Nonfarm Payroll Employment (PAYEMS).

For the purpose of this report, we use the data from January 1974 to December 2025 to construct the year-on-year employment growth rate (thus losing the first 12 observations). A summary of the two series is presented in Table 1.

Over the past five decades, the US labour market has experienced a sustained long-term growth, with monthly employment level going from 76.46 millions to a peak of 158.84 millions in August 2025. The average employment level during the period 1974-2025 has been around 120.8 millions. While the long-term trend is upward, the series exhibits clear cyclicity corresponding with the US business cycle, with dips in employment level during US recessions in 1982, 1991, 2001 (dot-com bubble), 2008-2009 (Great Financial Crisis) and 2020 (Covid-19 Crisis).

Meanwhile, the US employment growth rate appears to be fluctuating around a relatively stable level, with a mean of approximately 1.41%, and has turned negative only during recessions, without exhibiting a clear trend.

Table 1: Descriptive statistics for employment data

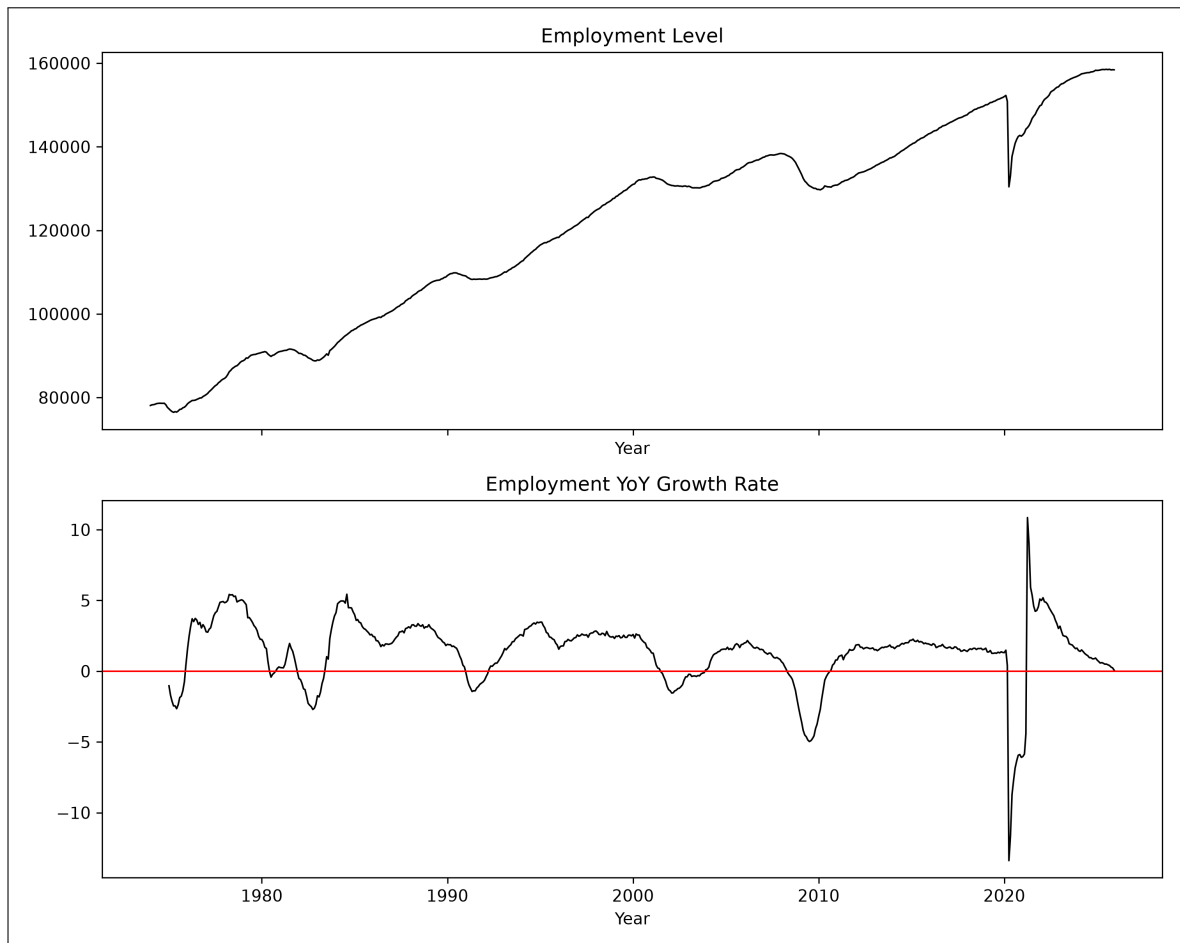
Variable	Mean	Median	Min	Max	St. dev
Employment	120805	129837	76460	158548	23205.18
Empl. Growth Rate	1.4161	1.6919	-13.3906	10.8575	2.29

Notes: *Employment* is the employment level observed each month, expressed in thousands of persons. *Empl. Growth Rate* is the change in employment level compared to the same month in the previous year and it is expressed in percentage points.

2.2 Stationarity

Figure 1 shows that the time series of employment shows a clear upward trend, with a significant drop during the Covid-19 pandemic in 2020 followed by a rapid recovery. Therefore, it is reasonable to conclude that the employment level does not appear to be covariance-stationary, since it follows a trend, and thus the series is not covariance-stationary. In contrast, the year-on-year employment growth rate time series appears to be stationary as it fluctuates around a constant mean, with no visible trend and relatively stable variance. The evidence is consistent with the employment level being non-stationary, while the year-on-year employment growth rate appears to be stationary. For this reason, we model the year-on-year growth rate rather than the employment level. In fact, modelling the employment level series could lead to misleading regression results, where a strong statistical relationship appears even if no true economic relationship exists. In contrast, the employment growth rate series allows for reliable estimation and inference, and ensure that shocks to employment have only temporary effects rather than permanent ones.

Figure 1: US employment level and YoY empl. growth rate (1974-2025)



Notes: The top chart shows a line plot of monthly employment level expressed in thousands of persons (Y-axis) against year (X-axis). The bottom chart shows a line plot of year-on-year employment growth rate expressed in percentage points (Y-axis) against year (X-Axis).

2.3 OLS Estimation

After ensuring that the employment growth rate series is stationary, we utilize the full sample (1975M1-2025M12) to estimate the model using OLS. Given the data on employment growth rate, the AR(3) can be written in matrix notation as follows:

$$\begin{bmatrix} y_{1975M4} \\ y_{1975M5} \\ \vdots \\ y_{2025M11} \\ y_{2025M12} \end{bmatrix}_{(612 \times 1)} = \begin{bmatrix} 1 & y_{1975M3} & y_{1975M2} & y_{1975M1} \\ 1 & y_{1975M4} & y_{1975M3} & y_{1975M2} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & y_{2025M10} & y_{2025M9} & y_{2025M8} \\ 1 & y_{2025M11} & y_{2025M10} & y_{2025M9} \end{bmatrix}_{(612 \times 4)} \begin{bmatrix} c \\ \phi_1 \\ \phi_2 \\ \phi_3 \end{bmatrix}_{(4 \times 1)} + \begin{bmatrix} u_{1975M4} \\ u_{1975M5} \\ \vdots \\ u_{2025M11} \\ u_{2025M12} \end{bmatrix}_{(612 \times 1)} \quad (10)$$

so it follows that the AR(3) in matrix notation can be written as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \quad (11)$$

After rewriting the model in this way, we can estimate the matrix $\boldsymbol{\beta}$ through OLS using the full sample. Note that the vector $\mathbf{y}$ begins from the fourth observation (y_{1975M4}) since we are dealing with an AR(3).

3 Results

3.1 Estimation results

From the OLS estimation we obtain the vector $\hat{\boldsymbol{\beta}}_{LS} = [\hat{c}, \hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3]$ containing the estimates of the parameters. The empirical results are presented in Table 2. All the estimates obtained are statistically significant at the 5% level, with $\hat{\phi}_1$ also significant at the 0.1% level. The estimated AR(3) model is:

$$\hat{y}_t = \underset{(0.04)}{0.11} + \underset{(0.04)}{1.01}y_{t-1} - \underset{(0.06)}{0.18}y_{t-2} + \underset{(0.04)}{0.09}y_{t-3} \quad (12)$$

The standard errors are reported in parentheses. The parameter ϕ_1 can be interpreted as the degree of persistence of the series due to the previous value. Specifically, a one percentage point increase in employment growth at time $t-1$ is associated with an increase of around 1.01 percentage points in employment growth at time t , holding the second and third lagged growth rates constant. This result suggests a high degree of persistence in the employment growth process, whereby periods of strong growth tend to be followed by similarly strong growth in subsequent periods. The coefficient ϕ_2 and ϕ_3 can be interpreted similarly as ϕ_1 .

To evaluate the stationarity of the estimated AR(3), we check the roots of the **characteristic equation**, which for the AR(3) is given by:

$$(1 - \phi_1 L - \phi_2 L^2 - \phi_3 L^3)y_t = c + u_t \quad (13)$$

By replacing the lag operator (L) by a variable z and setting the resulting polynomial equal to zero, we obtain the following equation:

$$1 - \phi_1 z - \phi_2 z^2 - \phi_3 z^3 = 0 \quad (14)$$

The values of z that solve Equation (11) are referred to as the **characteristic roots**.

In order for the process to be stationary, all the characteristic roots must lie outside the unit circle ($|z| > 1$). Using the estimated coefficients, the three characteristic roots were calculated numerically, yielding the solutions:

$$|z_1| = 1.086, \quad |z_2| = 3.198, \quad |z_3| = 3.198$$

Since all three roots satisfy $|z| > 1$, the stationarity condition for the estimated AR(3) process is satisfied.

Table 2: OLS estimates of the coefficients

Coefficient	Estimate	Std. Error	tstat	p-value
Lag 1 (ϕ_1)	1.01	0.04	25.01	0.0000
Lag 2 (ϕ_2)	-0.18	0.06	-3.15	0.0016
Lag 3 (ϕ_3)	0.09	0.04	2.34	0.0194
Intercept (c)	0.11	0.04	2.69	0.0073

Notes: The model is estimated via OLS using monthly data spanning from Jan. 1975 to Dec. 2025. The total number of observations is $N = 609$ (after accounting for the 3 initial lags used as regressors). t -stats are calculated against the null hypothesis $H_0 : \beta_i = 0$.

3.2 Companion form representation

Given the estimates of the coefficients in Table 2, we can rewrite the estimated model in companion form by replacing the companion matrices $\mathbf{c}$ and Φ with the estimates as follows:

$$\hat{\mathbf{c}} = \begin{bmatrix} 0.11 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{\Phi} = \begin{bmatrix} 1.01 & -0.18 & 0.09 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad (15)$$

The estimated model can be written in companion form as:

$$\begin{bmatrix} \hat{y}_{T+1|T} \\ y_T \\ y_{T-1} \end{bmatrix} = \begin{bmatrix} 0.11 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1.01 & -0.18 & 0.09 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_T \\ y_{T-1} \\ y_{T-2} \end{bmatrix}$$

Therefore, the estimate of the one-step ahead forecast is given by:

$$\hat{y}_{T+1|T} = 0.11 + 1.01y_T - 0.18y_{T-1} + 0.09y_{T-2} \quad (16)$$

Once the one-step ahead forecast is computed, the forecasts for the subsequent periods can be estimated through recursive substitution as shown in Equation (6). For our model, we use a forecast horizon of 24 months (see Figure 2 below).

3.3 Model diagnostic

After estimating the parameters, one can estimate the residuals as follows:

$$\hat{\mathbf{u}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} \quad (17)$$

Once we have estimated the residuals, it is possible to estimate the variance of the residuals to obtain a measure of the uncertainty associated with the model. The estimate of the residual variance is given by:

$$\hat{\sigma}^2 = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{T - K} = 0.76 \quad (18)$$

where T is the number of observations of $\mathbf{y}$ and K is the number of regressors. In our model, we obtained an estimate of residual variance equal to 0.76 (with $T - K = 609 - 4 = 605$). When computing forecasts, a commonly used measure of forecast uncertainty is the *Root Mean Squared Forecast Error (RMSFE)*, which captures the theoretical dispersion of forecast errors implied in the model. After calculating the estimate of the residual variance, one can calculate the RMSFE. The RMSFE for the one-step ahead forecast is equal to:

$$\text{RMSFE}(\hat{y}_{T+1|T}) = \sqrt{E[(y_{T+1} - \hat{y}_{T+1|T})^2]} = \sqrt{E(u_{T+1}^2)} = \sqrt{\sigma^2} = \sigma \quad (19)$$

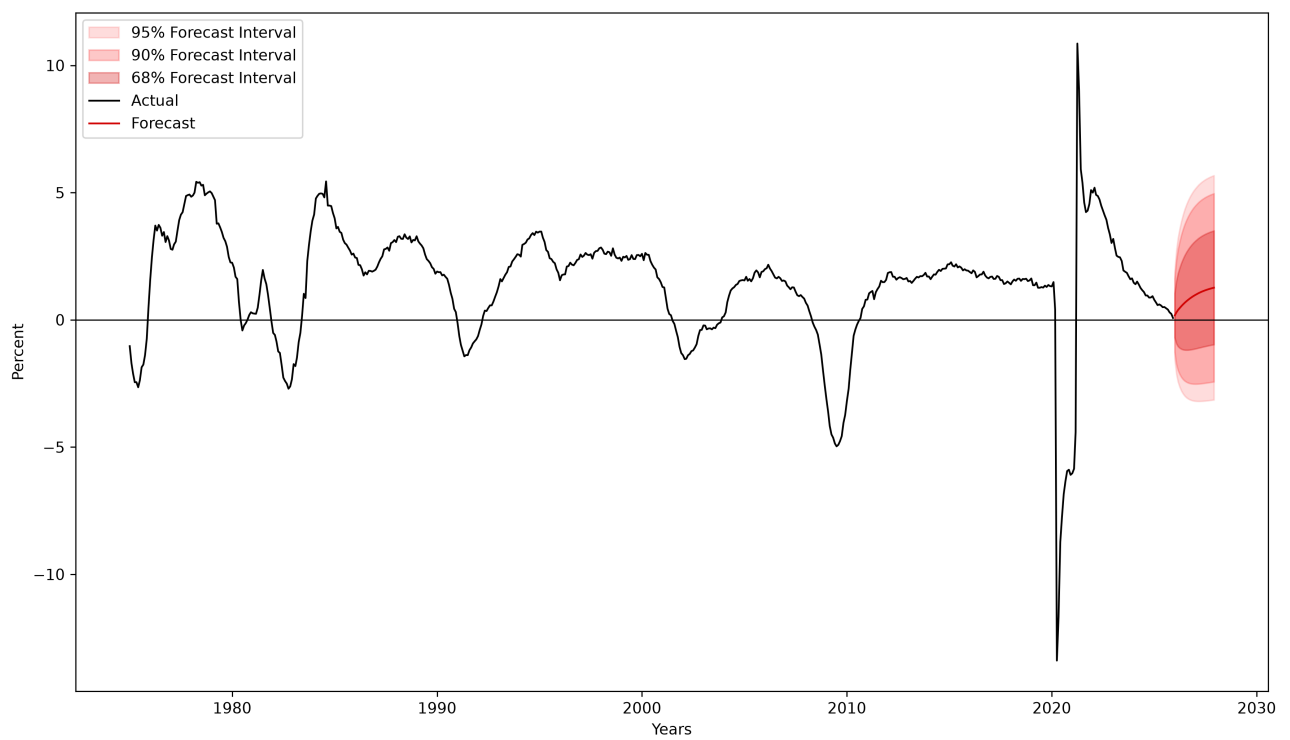
As the horizon of the forecasts increases, the RMSFE also increases because future forecast errors depend not only on the new shocks but also on the propagation of previous shocks through the structure of our model. For this reason, prediction intervals become wider as the forecast horizon increases. After calculating the RMSFE for the forecasted periods, it is possible to compute forecast intervals. For example, the 95% forecast interval is computed as:

$$\hat{y}_{T+h|T} \pm 1.96 \times \text{RMSFE}(\hat{y}_{T+h|T}) \quad (20)$$

3.4 Forecast

The point forecasts indicate a gradual increase in employment growth over the period January 2026 to December 2027. The point forecast starts at approximately 0% in January 2026 up to approximately 2% by the end of 2027. However, it should be noted that the forecast intervals include zero at all horizons, indicating substantial uncertainty around the point forecasts. The point forecast and the forecast intervals are presented in **Figure 2**.

Figure 2: Forecast of US YoY Employment Growth Rate (2026-2027)



Notes: 24-month ahead forecast for year-on-year percentage change of US employment level. Actual data for 1975M1-2025M12. Value in percent. Prediction interval at 68%, 90% and 95%.